

Lab report 13

1. Objective

The objective of this lab is to:

- Understand structures in C++
 - Learn how to declare structure types
 - Work with structure variables
 - Access structure members using dot operator
 - Understand input and output of structure data
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2. Theory

A structure in C++ is a user-defined data type that allows grouping of different data types under one name. Unlike arrays, structures can store multiple types of data such as integer, float, and character arrays together.

Structure members are accessed using the dot operator.

A structure declaration does not allocate memory; memory is allocated only when a structure variable is created.

3. Structure Concept

A structure can contain:

- Integer variables
- Float variables
- Character arrays

It is useful for representing real-world entities such as a part, person, or employee.

4. Program Examples

4.1 Structure with One Variable

```
#include <iostream>
using namespace std;

struct part
{
    int modelnumber; // yahan model number store kiya ja raha hai
    int partnumber; // yahan part number store kiya ja raha hai
    float cost; // yahan cost store ki ja rahi hai
};

int main()
{
    part part1; // yahan structure variable banaya gaya hai

    part1.modelnumber = 6244; // yahan model number assign kiya gaya hai
    part1.partnumber = 373; // yahan part number assign kiya gaya hai
    part1.cost = 217.55F; // yahan cost assign ki ja rahi hai

    cout << "Model: " << part1.modelnumber << endl; // yahan model display ho
    raha hai
    cout << "Part: " << part1.partnumber << endl; // yahan part display ho
    raha hai
    cout << "Cost: " << part1.cost << endl; // yahan cost display ho rahi hai

    return 0;
}
```

4.2 Structure with Two Variables

```
#include <iostream>
using namespace std;

struct part
{
    int modelnumber; // yahan model number store kiya ja raha hai
    int partnumber; // yahan part number store kiya ja raha hai
    float cost; // yahan cost store ki ja rahi hai
};

int main()
{
    part part1 = {6244, 373, 217.55F}; // yahan pehla structure initialize
    kiya gaya hai
    part part2; // yahan doosra structure banaya gaya hai

    cout << "First Structure Values" << endl; // yahan pehle structure ka
    output dikhaya ja raha hai
    cout << part1.modelnumber << endl; // yahan model display ho raha hai
    cout << part1.partnumber << endl; // yahan part display ho raha hai
    cout << part1.cost << endl; // yahan cost display ho rahi hai

    part2 = part1; // yahan structure copy kiya ja raha hai
}
```

```

        cout << "Second Structure Values" << endl; // yahan second structure ka
        output dikhaya ja raha hai
        cout << part2.modelnumber << endl; // yahan model display ho raha hai
        cout << part2.partnumber << endl; // yahan part display ho raha hai
        cout << part2.cost << endl; // yahan cost display ho rahi hai

        return 0;
    }

```

4.3 Employee Information Using Structure

```

#include <iostream>
using namespace std;

struct Person
{
    char name[50]; // yahan naam store kiya ja raha hai
    int age; // yahan age store ki ja rahi hai
    float salary; // yahan salary store ki ja rahi hai
};

int main()
{
    Person p1; // yahan structure variable banaya gaya hai

    cout << "Enter Full Name: "; // yahan user se naam liya ja raha hai
    cin.get(p1.name, 50); // yahan full name input liya ja raha hai

    cout << "Enter Age: "; // yahan age input li ja rahi hai
    cin >> p1.age; // yahan age store ki ja rahi hai

    cout << "Enter Salary: "; // yahan salary input li ja rahi hai
    cin >> p1.salary; // yahan salary store ki ja rahi hai

    cout << "\nDisplaying Information" << endl; // yahan output show kiya ja
    raha hai
    cout << "Name: " << p1.name << endl; // yahan name display ho raha hai
    cout << "Age: " << p1.age << endl; // yahan age display ho rahi hai
    cout << "Salary: " << p1.salary << endl; // yahan salary display ho rahi
    hai

    return 0;
}

```

5. Conclusion

In this lab, we learned:

- Structure declaration and usage in C++
- Structure variables and initialization

- Accessing members using dot operator
- Copying structure variables
- Input and output of structure data